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Polymer Chain Conformations in CNT/PS Nanocomposites from Small Angle Neutron Scattering

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Supporting Information

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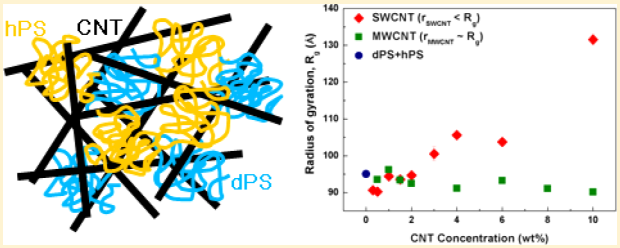
ABSTRACT: Chain conformations dictate many of the important physical properties of polymers including their dynamics. Using small angle neutron scattering, we probed chain conformations, specifically the radius of gyration (R_g), in both SWCNT/polystyrene ($r_{SWCNT}/R_g \sim 0.4$) and MWCNT/polystyrene ($r_{MWCNT}/R_g \sim 1$) nanocomposites. We fit the scattering data using a model that includes an ideal Gaussian chain to describe the polymer conformation and a rod network to describe the carbon nanotube (CNT) network. The scattering contribution from the rod network increases with the CNT concentration in both SWCNT and MWCNT nanocomposites, and the contribution is higher for SWCNT nanocomposites due to the smaller mesh size and higher mesh density. When the SWCNT and MWCNT concentrations are below 2 wt %, there is no significant change in R_g. Above 2 wt %, R_g in the MWCNT nanocomposites decreases slightly, while R_g in the SWCNT nanocomposites increases monotonically as a function of CNT concentration, showing a ~30% increase at 10 wt % SWCNT loading. Although we previously found a minimum in the tracer diffusion coefficient near the critical nanotube concentration, this trend is absent in the concentration-dependent polymer conformation.



1. INTRODUCTION

Polymer nanocomposites containing cylindrical, spherical, or layered nanoscale fillers can have enhanced properties including mechanical, electrical, and flammability.[1−3] However, the understanding of how nanofillers impact the structural and dynamic nature of polymer chains is still in its infancy. In this work, we investigate polymer chain conformations in the presence of the most widely studied cylindrical nanofiller, namely carbon nanotubes (CNTs).

Carbon nanotubes have many impressive properties including high mechanical strength, high electrical and thermal conductivities, and it has been shown that adding CNTs into various matrices (*e.g.*, polymers, ceramics, and metals) can dramatically improve the properties of the host materials[4] and different methods of making CNT/polymer nanocomposites have been reported.[5] Polymer diffusion[6−8] and polymer dynamics[9] have also been studied. The dispersion state of carbon nanotubes in polymer matrices is known to be a critical factor that determines the final properties of polymer nanocomposites.[10−15] When CNTs form aggregates or networks in solution[16] or a polymer matrix,[17] the aggregates and networks have been described as fractal objects via small-angle scattering experiments.[18] The dispersion state of CNTs in CNT/PS nanocomposites depends on processing methods, and the coagulation or rapid precipitation method for fabricating nanotube polymer nanocomposites has been shown to yield good CNT dispersions.[19,20] At low concentrations, the CNTs exist as isolated rods and small bundles, while at high concentrations the CNTs form low-density, electrically conductive networks in our nanocomposites. Good nanoparticle dispersion is particularly important when monitoring changes in the polymer conformation.[20]

The effect of nanofillers on polymer conformation, specifically the radius of gyration (R_g), has been previously studied for a variety of spherical nanoparticles. Nakatani et al. published the first experimental results describing a decrease for R_g in polysilicate/poly(dimethylsiloxane) nanocomposites when R_g is about the same size as the fillers and, conversely, an increase when R_g is larger than the filler.[21] Since then, the radius of gyration has been reported for a variety of spherical-nanofiller/polymer systems. Sen et al.[22] showed no change in polymer conformation for a wide range of R_g to spherical nanoparticles radius (r) ratios (1−3) and filler concentrations (up to 27.5 vol. %), but their TEM images suggest poor nanoparticle dispersion. Recently, Jouault et al.[23] reported a constant R_g in silica/polystyrene (PS) nanocomposites with $r/R_g < 1$,[18] but again the nanoparticle dispersion is less than ideal. Tuteja et al.[24] reported chain swelling when $r/R_g < 1$ in PS-nanoparticle/PS nanocomposites (soft particle system). In silica/poly(ethylene-propylene) nanocomposites, Nusser et

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al.[25] reported that R_g decreases when $r/R_g \sim 1$ and remains constant when $r/R_g > 1$ with increasing silica content. Their silica nanoparticles are grafted with hydrocarbon chains to improve the nanoparticle dispersion in a polymer matrix, although some aggregation is still observed at high loadings (>35 vol. %). In contrast, this paper studies polymer conformations in the presence of cylindrical nanoparticles, namely CNT/polymer nanocomposites.

Beyond experimental studies, simulations have been used to explore how polymer chain conformations are affected by varying the particle-to-polymer size ratio (r/R_g) and the interaction energy, but the results still have discrepancies. Most of the simulations study spherical particles. For larger nanoparticles with $r/R_g > 1$, most of the simulation work found no change in R_g.[26−28] For smaller nanoparticles with $r/R_g < 1$, Termonia found chain swelling in their Monte Carlo simulation,[26] while Vacatello found a constant R_g.[29] Hooper and Schweizer applied polymer reference site model (PRISM) to study the miscibility of nanoparticles and polymers and the local particle−polymer structure as a function of particle-to-monomer size ratio, degree of polymerization, strength (ε_{pc}) and spatial range of monomer-particle attractions and interfiller attractions.[30,31] They have shown that the miscibility windows of hard-sphere nanoparticles and polymer melts requires intermediate ε_{pc} (~0.5−2 k_BT). At low ε_{pc}, depletion attraction between nanoparticles is observed, resulting in phase separation; while at high values of ε_{pc}, adsorbed polymer chains bridge multiple nanoparticles forming a network phase. Other parameters have been investigated to shape the miscibility window. Using self-consistent PRISM, Frischknecht et al. showed an expansion of chain dimensions in nanoparticle/polymer systems with $r/R_g < 1$ and attractive interactions between polymers and fillers,[32] These simulation results were discussed in conjunction with experiments.[24] For the case of cylindrical nanoparticles, Karatrantos et al. investigated the structure and conformations of polymer chains in SWCNT/polystyrene (PS) nanocomposites, where R_g is larger than the radii of the SWCNTs.[33] Although the local chain structure is affected, they found R_g to be constant over a range of polymer-nanotube interaction energies and SWCNT radii. These simulations are the most analogous to our system and will be compared to our experimental results.

In this paper, we use small angle neutron scattering (SANS) to determine the R_g in CNT/PS nanocomposites. Our nanocomposites are prepared by a coagulation method, which has been previously shown to yield good dispersion of CNTs in a polymer matrix.[19] Moreover, the interaction energies between CNTs and PS are negligible.[8] The R_g of the matrix PS is larger than the radius of the SWCNTs and about the same as the radius of the MWCNTs, so as to explore both $r/R_g < 1$ and $r/R_g \approx 1$. Importantly, we experimentally determine and apply the contrast matching condition between CNTs and the PS matrix to minimize the scattering contribution from the CNTs.

2. EXPERIMENTAL METHODS

2.1. Materials. The nanocomposites used in this work are composed of single-walled carbon nanotubes (SWCNT), multiwalled carbon nanotubes (MWCNT), deuterated polystyrene (dPS), and hydrogenated polystyrene (hPS). SWCNTs synthesized by high-pressure carbon monoxide conversion (HiPco) were purchased from Unidym; MWCNTs produced by chemical vapor deposition (CVD) were purchased from Nanolab. SWCNTs and MWCNTs were purified by oxidizing in air at 150 °C for 24 h followed by an HCl treatment.[34] The dPS and hPS were synthesized via standard living anionic polymerization methods.[6] The weight-averaged molecular weight and PDI for dPS (116 kg/mol, 1.03) and hPS (117 kg/mol, 1.05) were measured by size exclusion chromatography (SEC) using a Viscotek TDA 302 with refractive index, viscosity and light scattering detectors (with a 690 nm wavelength laser).

2.2. Nanocomposites Preparation. All nanocomposites were fabricated by the coagulation method.[19] CNTs were well dispersed in dimethylformamide (DMF) by sonication for 24 h, and a mixture of dPS and hPS with the desired volume ratio were dissolved in DMF by stirring for 24 h. After mixing these two solutions, the CNT/dPS + hPS mixtures were rapidly precipitated in DI water. The precipitate was dried, annealed at 150 °C under vacuum for 24 h, and then hot pressed into thin circular disks (diameter ~25 mm; thickness ~0.17 mm). The size of SWCNTs and MWCNTs were measured by atomic force microscopy (AFM) and scanning electron microscopy (SEM), respectively, and the results are given in Table 1. Note that the CNTs used for size determination and for the SANS samples were sonicated for the same amount of time, so the size distributions reported in Table 1 are indicative of the CNTs in the polymer nanocomposites. For contrast matching experiments, the nanocomposites were made with 1 wt % SWCNT in nine polymer matrices with dPS/hPS ratios from 79/21 to 63/37. The contrast matching experiments with 1 wt % SWCNTs found the contrast matching condition is dPS/hPS = 0.725/0.275 (see section 4). Subsequently, the remaining CNT/dPS + hPS nanocomposites use this dPS/hPS ratio. For comparison a dPS + hPS homopolymer mixture dPS/hPS = 72.5/27.5 was prepared by the same coagulation procedure to provide the radius of gyration of PS in the absence of CNTs. Table 2 lists the details of the samples along with the location of the SANS experiments. In these nanocomposites, the CNTs are hollow, which implies that the scattering length density will depend on the radius of a SWCNT and on both the radius and number of layers in a MWCNT, thereby complicating the effort to define a contrast matched state.

Table 1. Characteristics of the Single-Walled and Multi-Walled Carbon Nanotube Bundles

	diameter (nm)	length (nm)	aspect ratio
SWCNT	7.4 ± 3.1	327 ± 142	~44
MWCNT	20 ± 6	650 ± 450	~33

Table 2. Sample Information for Contrast Matching Experiments and the Study of the Polymer Conformation in CNT/PS Nanocomposites with Different CNT Concentrations[a]

	CNT concentration (wt %)	dPS/hPS (v/v)	SANS
Contrast Matching Experiments			
SWCNT/ dPS + hPS	1	79/21, 77/23, 75/25, 73/27, 71/29, 69/31, 67/33, 65/35, 63/37	ORNL
Polymer Conformation Experiments			
dPS + hPS	0	72.5/27.5	NIST
SWCNT/ dPS + hPS	0.3, 0.5, 1, 1.5, 2, 4, 6, 10	72.5/27.5	ILL
MWCNT/ dPS + hPS	0.5, 1, 1.5, 2, 4, 6, 8, 10	72.5/27.5	ILL

[a]SANS experiments were performed at Oak Ridge National Laboratory (ORNL), Institute Laue-Langevin (ILL), and National Institute of Standards and Technology (NIST).

2.3. Small Angle Neutron Scattering (SANS). The differential scattering cross-section normalized by a unit volume, $(d\Sigma/d\Omega)(q)$, contains information about the size, shape, and interactions between the scattering centers. It is also called the macroscopic differential scattering cross-section, and has dimensions of inverse length, usually

cm^{-1}. Sometimes it is inaccurately referred to as the scattering intensity $I(q)$, which actually represents the number of neutrons scattered through a particular angle and arriving on a small area of the detector in a unit time and has the unit (no. of neutrons/s). The relationship between $I(q)$ and $(d\Sigma/d\Omega)(q)$ can be expressed as[35]

$$I(q) = I_0(\lambda)\Delta\Omega\eta(\lambda)T(\lambda)V_s\frac{\partial\Sigma}{\partial\Omega}(q) \quad (1)$$

The first three terms on the right side of eq 1 are instrument specific, where $I_0(\lambda)$ is the incident neutron flux (unit: no. of neutrons/s·cm^2), $\Delta\Omega$ is the solid angle element determined by the physical size and position of the detector, and $\eta(\lambda)$ represents the detector efficiency. The remaining terms are sample specific: $T(\lambda)$ is the neutron transmission of the sample, and V_s is the sample volume impinged by the neutron beam, and finally $(d\Sigma/d\Omega)(q)$ is the differential scattering cross-section normalized by a unit volume. To determine $(d\Sigma/d\Omega)(q)$ of a sample, we measure the scattering intensity from the sample of interest, an empty cell, the calibration sample, and background noise. For a fixed wavelength instrument, the $(d\Sigma/d\Omega)(q)$ of our sample can be calibrated in absolute units (cm^{-1}) according to[35]

$$\frac{d\Sigma}{d\Omega}(q)_{sa} = \frac{\left[\left(\frac{I(q)_{sa}}{count_{sa}} - \frac{I(q)_{ba}}{count_{ba}}\right) - \frac{T_{sa}}{T_{ec}} \times \left(\frac{I(q)_{ec}}{count_{ec}} - \frac{I(q)_{ba}}{count_{ba}}\right)\right]}{\left[\left(\frac{I(q)_{ca}}{count_{ca}} - \frac{I(q)_{ba}}{count_{ba}}\right) - \frac{T_{ca}}{T_{ec}} \times \left(\frac{I(q)_{ec}}{count_{ec}} - \frac{I(q)_{ba}}{count_{ba}}\right)\right]} \times \frac{L_{sa}{}^2 t_{ca} T_{ca}}{L_{ca}{}^2 t_{sa} T_{sa}} \times \frac{\partial\Sigma}{\partial\Omega}(q)_{ca} \quad (2)$$

Here for the subscripts, *ca*, *sa*, *ec*, *ba* refer to the calibration standard, sample, empty cell, and background, respectively. In addition, L is the sample-to-detector distance, t is the sample thickness, and T denotes the transmission. For simplicity, $i(q)$ will be used to represent $(d\Sigma/d\Omega)(q)$.

Samples were measured at three neutron scattering facilities; see Table 2 for details. The first experiments conducted found the contrast matching condition for the SWCNTs. Here, 1 wt % SWCNT with different dPS/hPS volume ratio polymer matrices were measured on the Bio-SANS instrument in the neutron science division of Oak Ridge National Laboratory in Oak Ridge, TN. Three sample−detector distances, L = 0.3 and 6 m for λ = 6 Å and L = 14.5 m for λ = 18 Å, provide a q-range of 0.0025−0.15 $Å^{-1}$. After obtaining the contrast matching conditions for SWCNTs, SANS measurement for SWCNT/PS and MWCNT/PS nanocomposites with a fixed dPS/hPS ratio and different CNT concentrations were performed on instrument D11 in the Institute Laue-Langevin in Grenoble, France. Three sample−detector distances with L = 1.2, 8, and 20 m for λ = 9.7 Å cover the q-range of 0.0018−0.3 $Å^{-1}$. It is assumed that the contrast matching condition for MWCNTs is comparable to that for SWCNTs. Finally, a homopolymer sample with dPS/hPS = 72.5/27.5, which is the contrast matched condition for SWCNTs, was measured on the NG3 instrument at the NIST center of neutron research in Gaithersburg, MD, USA. Three sample−detector distances with L = 1.3 and 4 m for λ = 6 Å and L = 13 m for λ = 8.4 Å, provided a q-range of 0.001−0.4 $Å^{-1}$.

2.4. Glass Transition Temperature. The glass transition temperatures (T_gs) of the CNT/dPS + hPS nanocomposites were measured using modulated-temperature differential scanning calorimetry (TA Instruments Q2000) to separate the recoverable and irrecoverable processes. A modulation amplitude of ±1 °C and a period of 60 s were used with a ramping rate of 2 °C/min to heat the samples to 150 °C. The T_gs were determined from the second order transition in the second heating curve. The T_gs of PS and CNT/PS nanocomposites are comparable, Figure 1, indicating that adding CNTs does not alter the average segmental motion of PS.

2.5. Electrical Measurements. The electrical conductivities of the CNT/dPS + hPS nanocomposites were measured to extract the critical concentrations for electrical percolation, which correspond to the point at which CNT networks form. After the SANS measurements a strip was cut from the nanocomposites and current as a function of applied voltage was measured using a Keithley 6517A electrometer with an integrated LabView program. The data were fit using the fluctuation-assisted tunneling model[36] which has been proven to describe the nonlinearities in the I−V curves for silver nanowire networks in polymer nanocomposite systems.[36,13] For a system with conducting regions separated by insulating barriers, the model, which is based on the field enhancement of electron tunneling probability, describes the voltage-dependent transition between low-field conductance $G_0(V\to 0)$ and high-field conductance $G_h(V\to\infty)$.

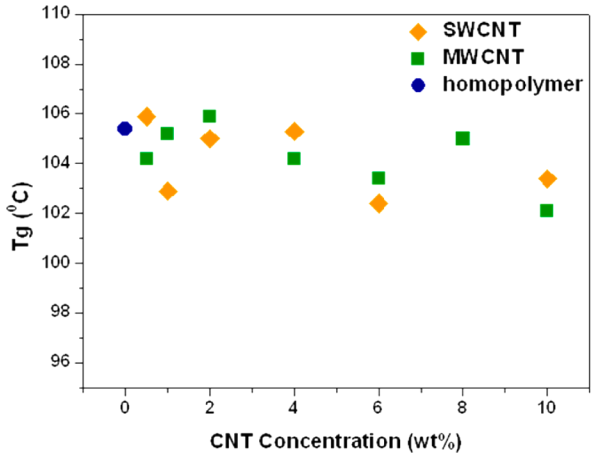

Figure 1. Modulated DSC results provide T_g as a function of CNT concentration. The width of the second order transition is ∼6−7 °C in all the samples.

$$G = \frac{I}{V} = \frac{G_o \exp(V/V_o)}{1 + (G_o/G_h)[\exp(V/V_o) - 1]} \quad (3)$$

The barrier voltage V_0 is related to the barrier height between the conducting regions, and gives an exponential increase in conductance as V increases. The electrical conductivity (σ) for the CNT/PS nanocomposites is $\sigma = G_h{}^*(w_s t_s/l_s)$, where w_s, t_s, and l_s are the sample width, thickness, and length, respectively; see Figure S1 in the Supporting Information. The composition-dependent conductivities were then fit to a power law with two fitting parameters (C, α)

$$\sigma = C\left(\frac{\phi - \phi_c}{\phi_c}\right)^{\alpha} \quad (4)$$

to obtain the critical concentration, ϕ_c, of these nanocomposites, Figure S2, Supporting Information. Note that eq 4 applies to concentrations above the percolation concentration and the same mass density was used for both SWCNTs and MWCNTs to convert weight fraction to volume fraction. The critical concentrations for these SWCNT/dPS + hPS and MWCNT/dPS + hPS nanocomposites are 0.47 vol. % and 1.45 vol. %, respectively. The lower percolation for the SWCNT nanocomposites is consistent with their higher aspect ratio.

3. CONTRAST MATCHING AND SCATTERING MODEL FOR CNT/POLYMER NANOCOMPOSITES

3.1. Contrast Matching CNTs and Polymer Matrix. The characteristic length scales of the CNT networks overlap with the length scales associated with the polymer conformations. To reduce the scattering contribution from the CNT networks, we contrast matched the CNTs to the mixture of dPS and hPS. Specifically, we tune the volume ratio of dPS to hPS to match the average scattering length density of the polymer matrix to that of the SWCNTs. Note that exact contrast matching was unachievable, so our fitting model described below will still account for the CNT network. The scattering intensity for a

three component system (*e.g.* SWCNT, dPS, and hPS) can be expressed as[37]

$$i(q) = [x(\rho_{dPS} - \rho_{swcnt})^2 + (1 - x)(\rho_{hPS} - \rho_{swcnt})^2]P(q) + [x\rho_{dPS} + (1 - x)\rho_{hPS} - \rho_{swcnt}]^2 Q(q) \quad (5)$$

where x is the volume fraction of dPS relative to the total volume of polymer, ρ_i is the scattering length density of dPS, hPS and SWCNT. $P(q)$ is proportional to product of the molar fraction of polymer chains Φ_{pol} and the form factor of a single Gaussian chain, which is dependent on the degree of polymerization. $Q(q)$ represents all other contributions to the total scattering intensity. The purpose of contrast matching is to minimize $Q(q)$ and thereby accentuate the scattering from the polymer chains. The contrast matched condition is

$$x\rho_{dPS} + (1 - x)\rho_{hPS} - \rho_{swcnt} = 0 \quad (6)$$

The scattering length density of SWCNT was calculated for λ = 10 Å using the scattering length density calculator provided by NIST[38] and a SWCNT mass density 1.5 g/cm^3: ρ_{swcnt} = ~ 5.0 × 10^{10} cm^{-2}. A single monomeric unit was taken as the smallest scattering unit (C_8D_8 for dPS and C_8H_8 for hPS) and using the scattering lengths of deuterium, hydrogen, and carbon, and the density of dPS and hPS as 1.13 g/cm^3 and 1.04 g/cm^3, we found ρ_{dPS} = 6.47 × 10^{10} cm^{-2} and ρ_{hPS} = 1.42 × 10^{10} cm^{-2}. Using eq 6 we estimated the contrast matching condition in SWCNT/dPS + hPS nanocomposites to have a dPS/hPS ratio of 71/29. Thus, nanocomposites with dPS/hPS volume ratios from 63/37 to 79/21 were prepared.

3.2. Scattering Model. Our scattering model for CNT/dPS + hPS nanocomposites considers four contributions to the total scattering intensity, namely the polymer chains, the CNT network, voids in the sample, and incoherent scattering. The polymer chains are described as ideal Gaussian chains and the scattering intensity is[37]

$$i(q)_{poly} = V\Phi_{poly}\Delta\rho^2 P(q)_{poly} = V\Phi_{poly}\Delta\rho^2 \frac{1}{N^2}\left[N\frac{1 + b}{1 - b} - 2b\frac{1 - b^N}{(1 - b)^2}\right] \quad (7)$$

$$b = \exp(-l_k^2 q^2/6)$$

where $P(q)_{poly}$ is the single chain form factor, N is the number of Kuhn monomers, l_k is the Kuhn length, V is the volume of a polymer chain, Φ_{poly} is the polymer volume fraction, and $\Delta\rho$ is the contrast between the corresponding scattering length densities of the polymer chains and the matrix. For large N, $P(q)_{poly}$ can be simplified to the Debye function[37,39]

$$\mathrm{Debye}(q, R_g) = \frac{2}{(qR_g)^4}[q^2R_g^2 - 1 + \exp(-q^2R_g^2)] \quad (8)$$

where R_g is the radius of gyration of the polymer chain. Therefore, the first term in our scattering model is A × Debye(q, R_g), where A is the prefactor of the Debye function related to the scattering length densities of the different species, monomer volumes, the degree of polymerization, and the volume ratio of the polymers. In miscible polymer blends where the polymer−polymer interactions might perturb the polymer conformation, the scattering intensity is given by the random phase approximation.[37] However, given that the Flory−Huggins parameter for dPS and hPS is ~10^{-4} at 160 °C,[40] we neglect the effect of interactions and use just one Debye function to describe a Gaussian conformation in the fitting model.

The nanocomposites contain CNT networks that can be described as mass fractal objects, a concept first introduced by Mandelbrot.[41] A fractal object has a self-similar structure and its mass can be described by

$$M(R) \sim (R/r_0)^D \quad (9)$$

where R is a linear dimension, r_0 is the gauge of measurement, and D is the fractal dimension. The structure factor, $S(q)$, for a fractal rod-like network has previously been derived[42]

$$S(q, D, \xi, r_0) = 1 + \frac{D\exp(\Gamma(D - 1))\sin(D - 1)\tan^{-1}(q\xi)}{(qr_0)^D[1 + (q\xi)^{-2}]^{(D-1)/2}} \quad (10)$$

where $\Gamma(x)$ is the gamma function and ξ is the characteristic length of the fractal object above which the mass distribution of the object can no longer be described as fractal. The scattering intensity, $i(q)$, is proportional to the product of the structure factor, $S(q)$, and the form factor, $P(q)$. In CNT nanocomposites, $S(q)$ describes the scattering from CNT networks (eq 10), and $P(q)$ is the form factor for CNTs. Similar approaches have been used previously to describe the scattering intensity for CNTs in solutions or polymers.[17,43,44] Figure 2 illustrates that the characteristic length scales accessed by SANS and associated with the CNT network can be larger or smaller than the diameters of CNT bundles. Thus, at modest scattering angles ($\xi^{-1} \leq q \leq r_0^{-1}$: the green circle indicates that the probing length scale is larger than the average mesh size, s) the scattering intensity is dominated by the network structure factor and reduces to $S(q) \sim q^{-D}$. Previous work has reported that $D \sim 2$ for well dispersed rod networks.[16,45] At higher scattering angles the second term in eq 10 is negligible, so $S(q) \sim 1$ and the $i(q)$ is dominated by $P(q)$, which for rods $\sim q^{-1}$. Specifically, for $P(q)$ to dominate the length scales being probed by the scattering experiment, q^{-1} must be smaller than the mesh size of the rod networks ($q^{-1} \ll s$: red circle). In these experiments this high q-range is inaccessible or overlapped with the region dominated by the polymer chain scattering. Thus, the scattering model accounts for the CNT network with a term that is proportional to q^{-2}, specifically $B \times q^{-2}$.

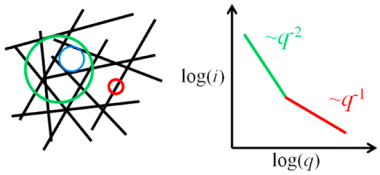

Figure 2. Schematic showing the relationship between the length scales probed (left) and the corresponding q regime along with the angular dependence of the scattering. When the probing length scale is greater (green) and smaller (red) than the mesh size (blue) formed by the CNTs, the expected angular dependencies are q^{-2} and q^{-1}, respectively.

At very low angle scattering angles, the scattering intensity is dominated by voids and defects in samples, and this scattering intensity is given by Porod's law[46]

$$i(q) = \frac{2\pi(\Delta\rho)^2 S}{q^4} \quad (11)$$

where $\Delta\rho$ is the difference between the scattering length densities of the scattering objects (voids, *etc.*) and the matrix, and S is the total area of the boundaries.[39] Notice that the scattering cross-section shown in eq 11 is not normalized by a unit volume, so its unit is cm^2. This contribution to the scattering model is $C \times q^{-4}$. Lastly, incoherent scattering, mainly from hydrogen, provides no structural information (q-independent) and this background intensity is incorporated via a constant, D. The four contributions (polymer, CNT network, defects, incoherent scattering) combine to give the following scattering model,

$$i(q) = A \times \text{Debye}(q, R_g) + B \times q^{-2} + C \times q^{-4} + D \tag{12}$$

When fitting the SANS data, we found that including the constant D only slightly improved the coefficient of determination (COD) and had minimal impact on the other parameters. Therefore, we omitted the incoherent background contribution in the scattering model,

$$i(q) = A \times \text{Debye}(q, R_g) + B \times q^{-2} + C \times q^{-4} \tag{13}$$

This model incorporates three scattering objects. From high q to low q these objects are the polymer chain with $R_g \sim 9$ nm, the mesh size of the nanotube network at tens to hundreds of nm, and defects and voids larger than ~ 1 μm.

3.3. Method for Fitting Scattering Model to SANS Data. First, we investigate the dimensionality of the CNT network by analyzing the SANS data from the contrast matching samples with 1 wt % SWCNT and nine dPS/hPS volume ratios using

$$i(q) = A \times \text{Debye}(q, R_g) + B \times q^{-n} \tag{14}$$

The best fits of eq 14 to the data find that the exponent in the second term is $n \sim 2$. This result confirms a rod-like network in these CNT/dPS + hPS nanocomposites. Subsequently, we fit eq 13 to the SANS data sets to obtain all four fitting parameters (A, B, C, and R_g). We set R_g as a shared fitting parameter, which means that R_g is forced to be the same for every data set during the fitting process, because R_g should not be affected by the dPS/hPS volume ratio. The other parameters (A, B, and C) are related to the volume ratio of dPS in the sample, so they are free fitting parameters for the nine dPS/hPS ratios. For the dPS + hPS homopolymer sample, $B = 0$, because there are no CNT networks.

For CNT/dPS + hPS nanocomposites with a dPS/hPS volume ratio of 72.5/27.5, we adapt eq 13 to account for the CNT concentration. The fitting parameter A is proportional to the volume fraction of polymer chains, so A is replaced by A' $(1 - \Phi_{cnt})$ to give

$$i(q) = A'(1 - \Phi_{cnt}) \times \text{Debye}(q, R_g) + B \times q^{-2} + C \times q^{-4} \tag{15}$$

where Φ_{cnt} denotes the volume fraction of CNTs. For SANS data from polymer nanocomposites with a fixed dPS/hPS ratio and different CNT concentrations (SWCNTs and MWCNTs are treated separately), we set A' as a shared fitting parameter, because A' is not affected by the CNT concentration. In addition, we set Φ_{cnt} as a fixed parameter for each nanocomposite. The remaining fitting parameters (R_g, B, and C) are free fitting parameters for each CNT concentration.

4. RESULTS AND DISCUSSION

4.1. Homopolymer. The homopolymer sample of dPS + hPS (72.5/27.5) was measured at NCNR, and the $i(q)$ data reduction was done in IGOR Pro.[47] Figure 3 shows the scattering intensity as a function of q and the associated fitting with eq 13 in Origin 8.6 with $B = 0$, because there is no CNT network. The shoulder in the scattering intensity is nicely captured by the Debye function and in the very high q region (>0.02 $Å^{-1}$), the scattering curve shows a q^{-2} dependence according to the Debye function. In the low q region, the upturn in the scattering intensity is fit by the q^{-4} term in our scattering model (eq 13). The R_g in the dPS + hPS 72.5/27.5 mixture is 9.50 ± 0.03 nm, which is close to the literature value of 8.8 nm (the difference is within 8%).[40]

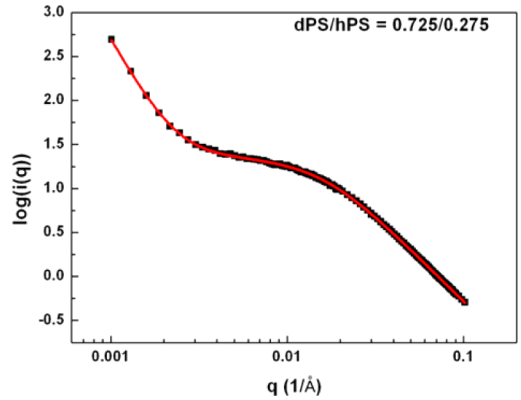

Figure 3. SANS scattering intensity for dPS/hPS homopolymer mixture with a volume ratio of dPS/hPS = 72.5/27.5. Equation 13 is fit to the data using $B = 0$. The low q upturn is described by the q^{-4} term (defect term) and the shoulder at $q \sim 0.01$ $Å^{-1}$ is captured by the Debye function with $R_g = 9.50 \pm 0.03$ nm.

4.2. Contrast Matching in 1 wt % SWCNT/dPS + hPS Nanocomposites. For the contrast matching experiments, samples with the same SWCNT concentration but different dPS/hPS volume ratios were measured at ORNL. The $i(q)$ data reduction was done using IGOR PRO and eq 13 fit to the reduced $i(q)$ data using Origin 8.6. Figure 4 shows that the

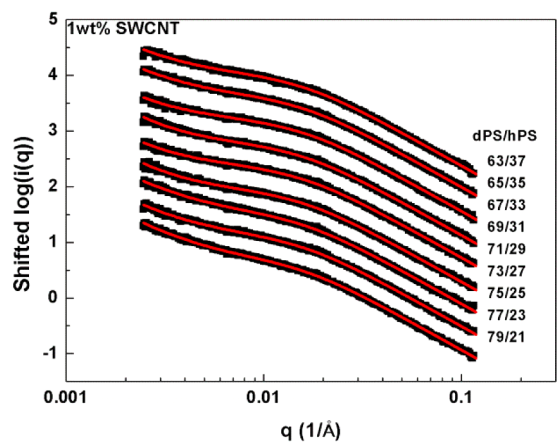

Figure 4. SANS scattering data for the contrast matching samples with 1 wt % SWCNT and a range of dPS/hPS volume ratios. Equation 13 (solid line) is fit to these data with R_g as a shared parameter and $R_g = 9.69 \pm 0.02$ nm. The data are shifted for clarity.

scattering intensity exhibits a shoulder and a low angle upturn. Our scattering model fits all the data well with a single value of $R_g = 9.69 \pm 0.02$ nm, which is only 2% higher than the R_g for the homopolymer.

Figure 5 shows the fitting parameters A and B used in Figure 4 to fit eq 13 as a function of dPS/hPS volume ratio (x). The parameter B is greater than zero at all dPS/hPS ratios, nearly constant from $x \sim 0.67-0.73$ and quite small across this range of x. This indicates that these dPS volume fractions are close to the ideal contrast matching condition, although perfect contrast matching was not achieved. By comparing eq 5 and eq 13 we find

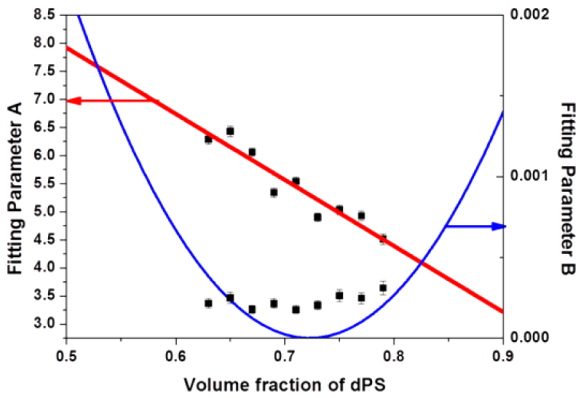

Figure 5. Plot of fitting parameters A and B (solid symbols) used in Figure 4 as a function of dPS volume fraction in the matrix. The parameter B is nearly constant for $x \sim 0.67-0.73$. Although B is very small, $B > 0$ implies that perfect contrast matching is unachievable. Equation 16 (red line) and eq 17 (blue line) were simultaneously fit. Equation 17 has a minimum at $x = 0.725$ that is selected as the contrast matching condition for subsequent nanocomposites.

$$A = A_o[x(\rho_{dPS} - \rho_{swcnt})^2 + (1 - x)(\rho_{hPS} - \rho_{swcnt})^2] \quad (16)$$

$$B = B_o[x\rho_{dPS} + (1 - x)\rho_{hPS} - \rho_{swcnt}]^2 \quad (17)$$

where $\rho_{dPS} = 6.47 \times 10^{10}$ cm^{-2} and $\rho_{hPS} = 1.42 \times 10^{10}$ cm^{-2} are known.[37] Figure 5 includes eq 16 and eq 17 simultaneously fit using $A_0 = 1.03$, $B_0 = 1.7 \times 10^{-3}$, and $\rho_{swcnt} = 5.06 \times 10^{10}$ cm^{-2}. As expected, A depends linearly on x. Moreover, the value of the scattering length density for SWCNT is very close (~1%) to the value we obtained from the scattering density calculator.[38] The minimum in eq 17 is used as the best contrasting matching condition between SWCNTs and the polymer matrix, namely $x = 0.725$. Therefore, we used a dPS/hPS volume ratio equal to 72.5/27.5 to prepare all our samples with different CNT concentrations.

4.3. CNT/dPS + hPS Nanocomposites as a Function of CNT Concentration. All SWCNT/dPS + hPS and MWCNT/dPS + hPS nanocomposites with the same dPS/hPS volume ratio and different CNT concentrations were measured at ILL and the data reduction performed with LAMP, which is developed by ILL. Figure 6 shows the SANS results for the SWCNT/dPS + hPS and MWCNT/dPS + hPS nanocomposites with 72.5/27.5 dPS/hPS volume ratio across a wide CNT-concentration range.

Figure 7 separates the various contributions to the scattering model for the MWCNT/dPS + hPS nanocomposites with the lowest and highest CNT loadings. The shoulder in the SANS data is modeled by the Debye function, which uses Gaussian chain statistics to describe the polymer conformation. The upturn at $q \sim 0.004-0.02$ Å^{-1} is captured by the rod network term (Bq^{-2}). At the lowest q range ($q < 0.004$ Å^{-1}), the rod network term is insufficient and the defect term is necessary, which is described by Cq^{-4}. At both low and high CNT concentrations, the combination of the three terms in eq 15 is necessary to fit the experimental SANS data. At higher MWCNT concentrations, as expected, the contribution from the rod network is greater than in the lower concentration.

In Figure 6, eq 15 is globally fit using Origin 8.6 to the reduced SANS data for the SWCNT nanocomposites with A' set as a shared variable; similarly the results from MWCNT nanocomposites were fit using a shared A' parameter. Equation 15, which uses the volume fraction of nanotubes, Φ_{cnt}, provides very good fits to all the SANS results. The fitting parameter C is quite small in all cases ($<1.1 \times 10^{-8}$) and does not exhibit a monotonic trend with CNT concentration (as expected), see Supporting Information. The fitting parameters for B and R_g will now be discussed separately.

As expected, the parameter B, which is the prefactor for the CNT network term in eq 15, increases with the CNT concentration, Figure 8. Interestingly, at a fixed CNT concentration B is larger for SWCNT nanocomposites than for MWCNT nanocomposites. To assess the observed

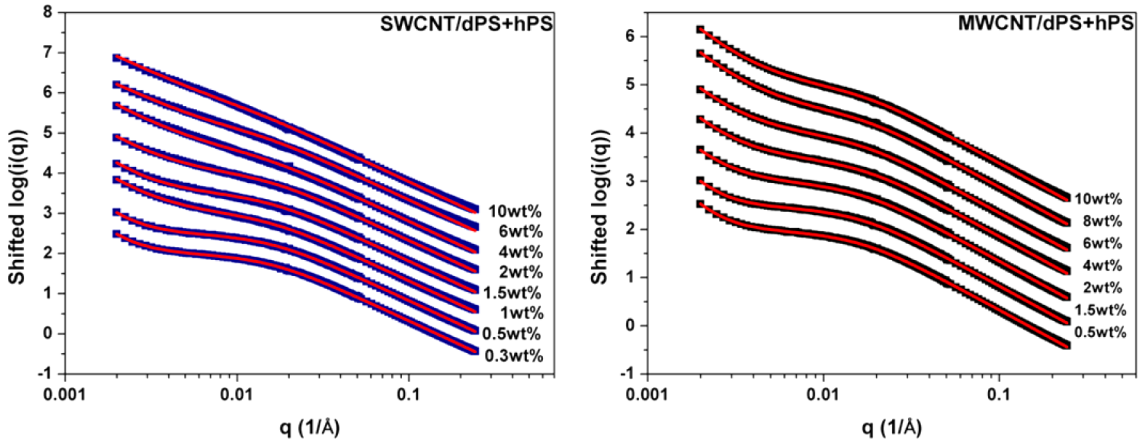

Figure 6. SANS data (points) for SWCNT/dPS + hPS and MWCNT/dPS + hPS nanocomposites with the 72.5/27.5 dPS/hPS volume ratio and different CNT concentrations. Equation 15 (red lines) is overlaid for each data set and data are shifted for clarity.

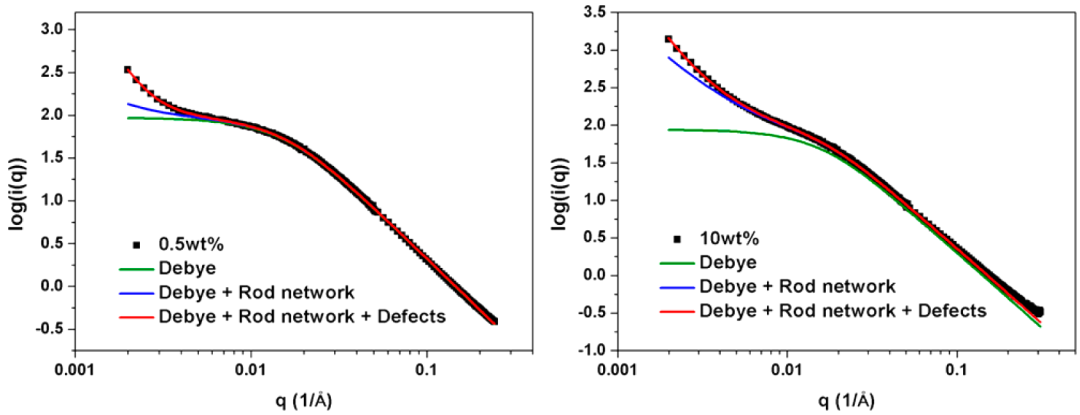

Figure 7. Fitting examples for the lowest (0.5 wt %) and the highest (10 wt %) CNT loadings for MWCNT/dPS + hPS (72.5/27.5) nanocomposites. The fitting curves broken into summations of different terms in eq 15 are indicated in the legend. All three contributions are needed to represent the data and, as expected, the rod network gives a larger contribution at the higher MWCNT concentration.

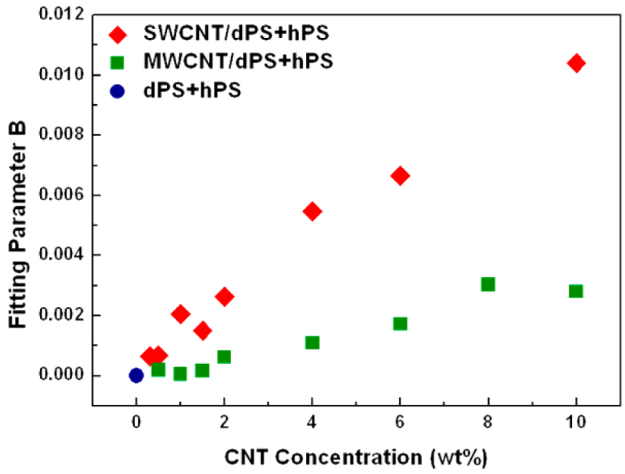

Figure 8. Prefactor B for the rod network term (q^{-2}) determined by fitting eq 15 to the SANS data in Figure 6 for SWCNT/dPS + hPS (red diamond) and MWCNT/dPS + hPS (green square) nanocomposites. For homopolymers (blue circle), B is zero.

difference in B in nanocomposites containing MWCNTs and SWCNTs we consider the mesh size as a function of both particle concentration and size. The probability distribution of mesh sizes for randomly and isotropically dispersed fibers in three dimensions is given by[48]

$$\frac{dP}{ds} = (4\pi\nu Ls + 4\pi\nu s^2)\exp\left[-\left(2\pi\nu Ls^2 + \frac{4\pi}{3}\nu s^3\right)\right] \quad (18)$$

where ν is the number density of fibers (1/nm^3), L is the half length of the fiber, and s represents the mesh size. For comparison to our experiments the half length (L) is half the average CNT length (Table 1) and the length, diameter and mass density were used to convert the weight percent of CNT to a number density of fibers (ν). Figure 9a shows the probability distribution of mesh sizes (eq 18) for nanocomposites with 0.5, 2, 6, and 10 wt % SWCNT and 10 wt % MWCNT. The most probable mesh size decreases with increasing SWCNT content and at a fixed CNT concentration (10 wt %) the SWCNT nanocomposite has a smaller average mesh size, because the SWCNTs have a higher aspect ratio and are smaller. The higher aspect ratio of the SWNCTs is also evident in the lower critical concentration for electrical percolation found for the SWCNT nanocomposites relative to the MWCNT nanocomposites, as detailed in section 2.5.

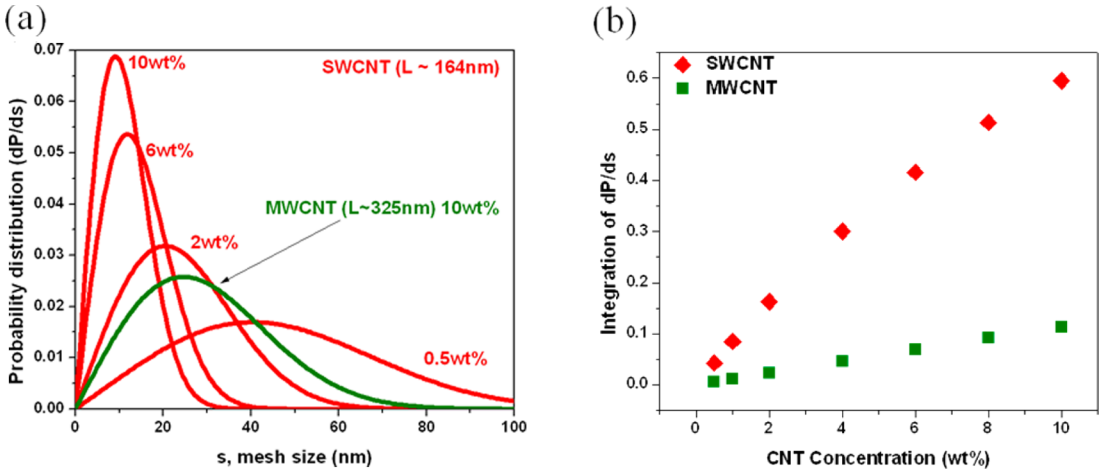

Figure 9. (a) Probability distribution of mesh sizes for SWCNT at different concentrations and 10 wt % MWCNT, calculated from eq 18. (b) Integration of dP/ds from s = 0 to 12 nm (shaded area) for SWCNTs and MWCNTs to correspond with the size range probed by the SANS experiments.

While eq 18 and Figure 9a describe the full size distribution of the CNT mesh (s = 0 to ∞), the SANS experiments probe only a finite range of mesh sizes. Specifically, the contribution from the CNT network is most evident from $q \sim 0.004$ to 0.02 Å^{-1}, corresponding to mesh sizes s from ~2 to 12 nm. Thus, to compare the fitting parameter B, which is proportional to the mesh concentration only within the length-scale probed by SANS, we integrate dP/ds for both SWCNTs and MWCNTs from s = 0 to s = 12 nm, Figure 9b. First, both the fitting parameter B and the integrated probability (s = 0−12 nm) increase with CNT concentration. Second, the values for SWCNT nanocomposites are higher than for MWCNT nanocomposites, which is mainly due the higher number density of SWCNTs. The rate of increase in B is ~3.6 times greater for SWCNTs than for MWCNTs (Figure 8). Similarly, the rate of increase for the integrated probability is ~5.6 times greater for SWCNTs (Figure 9b). Given the assumptions associated with eq 18 (monodisperse, straight fibers), this agreement is quite satisfying. This analysis provides valuable confirmation that at a fixed wt % CNT the SWCNT mesh is smaller in size and gives a larger contribution to the scattering. These conclusions are also qualitatively evident in the SANS plots for SWCNT/dPS + hPS and MWCNT/dPS + hPS (Figure 6), where the scattering contribution of the nanotube network is more pronounced in the SWCNT nanocomposites. Moreover, the shoulder in the SANS data that is associated with polymer chain scattering remains quite evident at the highest concentration in the MWCNT nanocomposites, while the contribution from the Debye function is subtle in the SWCNT nanocomposites due to the greater contribution of the CNT network. Overall, the agreement between Figure 8 and Figure 9b further confirms the appropriateness of our scattering model that includes a contribution from the rod network.

4.4. Radius of Gyration in CNT/dPS + hPS Nanocomposites. Figure 10 shows the R_g values extracted from the SANS data in Figure 6 using eq 15. When the concentration is below 2 wt %, R_g is approximately constant suggesting that R_g is unaffected at low SWCNT or MWCNT concentrations. Above 2 wt %, R_g in the MWCNT/dPS + hPS nanocomposites appears to slightly decrease. However, R_g increases significantly in SWCNT/dPS + hPS nanocomposites as the nanotube concentration increases. At 10 wt % SWCNT (i.e., highest concentration), R_g is ~13 nm. Relative to the homopolymer (0 wt % CNTs), this R_g represents a dramatic increase of 36%.

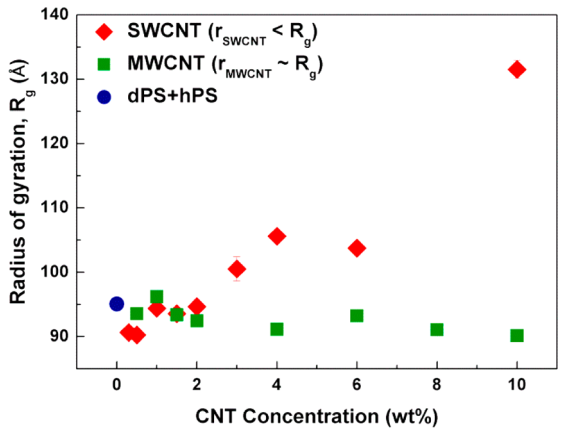

Figure 10. R_g for SWCNT/dPS + hPS and MWCNT/dPS + hPS nanocomposites versus CNT concentration. R_g is determined by fitting eq 15 to the scattering data in Figure 6. For comparison, the R_g value for dPS/hPS (72.5/27.5) is included (*cf.* Figure 3). Error bars are directly from fitting.

In Karatrantos' molecular dynamics simulation work,[33] they studied the change in polymer structure as a function of r_{swcnt}/R_g size ratios (0.08−0.59) and CNT polymer interactions (interaction strength between polymer chains and SWCNT: 1 to 5 k_BT) and found no change in R_g. In the simulations, the highest CNT loading is only about 0.8 vol. % and the system is represented by polymer chains surrounding an isolated SWCNT. Nevertheless, the SANS results agree with simulations, because no significant change in R_g is observed for concentrations less than 1.6 vol. % (2 wt %). Moreover, the temperature dependence of the diffusion coefficient in MWCNT/hPS nanocomposites[8] indicates only a very weak (or no) interaction between CNTs and polystyrene. This suggests that the R_g change we observe by SANS for SWCNT is not caused by the adsorption of polymer chains on SWCNT. Note that others have found that R_g increases as nanoparticle concentration increases for hard and soft *spherical* nanoparticles with r/R_g < 1.[21,24] For hard spherical nanoparticles with attractive particle−polymer interactions, Frischknecht et al. used self-consistent PRISM simulations to show that R_g increases as the nanoparticle concentration increases.[32] Conversely, when $r/R_g \sim 1$, R_g decreases slightly with increasing concentration in both our MWCNT nanocomposites and spherical nanoparticles at high loading.[25]

Previously we reported a diffusion minimum for polymer tracer diffusion in both SWCNT/PS and MWCNT/PS nanocomposites.[6−8] This behavior occurs when the polymer R_g is greater than the radius of the carbon nanotubes (r/R_g < 1), and the minimum occurs near the rheological percolation threshold (~0.5 wt % for SWCNT and ~2 wt % for MWCNT). From the reptation model, it has been shown that the diffusion coefficient is proportional to R^2/τ,[49] where R is the polymer chain size and τ is the relaxation time for a polymer chain to disentangle from its original tube. If we assume that τ is unaffected by the addition of CNTs into polystyrene, we expect a similar trend in R_g when we increase the CNT concentration in both SWCNT/PS and MWCNT/PS nanocomposites. The static properties of polymer chains we report in this work are difficult to reconcile with the trend of the tracer diffusion we obtained when increasing CNT concentration, which implies that other mechanisms are responsible for the observed minimum in the melt diffusion coefficient. More experiments, possibly quasi-elastic neutron scattering (QENS), nuclear magnetic resonance (NMR), and neutron spin echo that probe the dynamics of polymer segments at short time scales might provide more insight into the molecular mechanism or mechanisms that lead to a minimum in the polymer diffusion coefficient with increase CNT content.

5. CONCLUSION

We have set the foundation for studying polymer conformations in the presence of CNTs, and more generally in cylindrical nanoparticles. We used SANS to probe the polymer chain conformation in CNT/polymer nanocomposites as a function of nanotube concentration. Our scattering model incorporates four scattering features, namely polymer chains, rod networks, defects, and incoherent scattering, although the contribution from incoherent scattering was negligible. Contrast matching experiments identified the dPS/hPS volume ratio that minimizes the scattering contribution from CNTs,

although the rod network still makes a contribution. A Debye function fit the shoulder in the SANS data, indicating that polymer chains obey Gaussian chain statistics. Our observation agrees with recent simulations[33] indicating that R_g does not change appreciably at low CNT concentrations (<2 wt %). For MWCNT/polymer nanocomposites ($r/R_g \sim 1$), R_g decreases slightly as CNT concentration increases. In SWCNT/polymer nanocomposites ($r/R_g \sim 0.4$) above 2 wt %, R_g increases strongly by up to 36% as the SWCNT concentration increases.

The scattering contribution from the rod network term (B) increases as CNT concentration increases, as expected. Moreover, nanocomposites with SWCNTs have higher values for B than nanocomposites with MWCNTs at the same CNT concentration, because SWCNTs have a higher probability of smaller mesh size at the same mass concentration, which is due to the much higher number density for SWCNTs compared to MWCNTs. This is consistent with predictions of mesh size for networks of randomly and isotropically dispersed rods.

Future SANS studies could combine MWNCTs with higher molecular weight polymers to access the $r/R_g < 1$ regime and further explore the effect of relative size. Also, SANS studies of SWCNT/PS nanocomposites with aligned SWCNTs might reveal anisotropy in the polymer conformation relative to the nanotubes. Finally, selecting other cylindrical fillers (oxide or metal nanowires) could tune the polymer−filler interactions and possibly impact the polymer conformation.

■ ASSOCIATED CONTENT

Supporting Information

All the data for $I-V$ measurements and the associated fitting, which show that the critical (percolation) concentrations for SWCNTs and MWCNTs are ∼0.47 and 1.45 vol %, respectively, and values for the fitting parameter C as a function of CNT concentration. This material is available free of charge via the Internet at http://pubs.acs.org/.

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Notes

The authors declare no competing financial interest.

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